



EAST WEST UNIVERSITY
Department of Mathematics and Physical Sciences (MPS)
Semester: Summer 2025

Assignment-01

Course Code: MAT102: Differential Equations and Special Functions

Group No.	Sl. No.	Questions
01	1.	Form the differential equation by eliminating arbitrary constant(s) of the equation: $cx^2 + x + y^2 = 0$.
	2.	Obtain the differential equation of the family of straight lines through the fixed point (h, k) ; h and k not to be eliminated.
	3.	Reduce the following differential equation to a variables separable equation and hence solve it: $(\sqrt{x+y+1})\frac{dy}{dx} = 1$.
	4.	Solve the initial value problems (IVP): $2x(y+1)dx + y dy = 0$; when $x = 0$ and $y = 2$.
	5.	Solve the D.E.: (i) $(x^2 + y^2)dx + 2xydy = 0$ (ii) $[x \cos^2(y/x) - y]dx + x dy = 0$; when $x = 1, y = \pi/4$. (iii) $(2x^3 - xy^2 - 2y + 3)dx - (x^2y + 2x)dy = 0$. (iv) $(\tan y - 3x^2)dx + x \sec^2 y dy = 0$. (v) $\frac{dy}{dx} + y \sec^2 x = \sin x \sec^3 x$. (vi) $xy^2 \left(x \frac{dy}{dx} + y \right) = a^2$.
02	1.	Form the differential equation by eliminating arbitrary constant(s) of the equation: $y = cx + c^2 + 1$.
	2.	Obtain the differential equation of the family of straight lines with slope and x -intercept equal.
	3.	Reduce the following differential equation to a variables separable equation and hence solve it: $(x + y + 1)\frac{dy}{dx} = 1$.
	4.	Solve the initial value problems (IVP): $(1 + y^2)dx - (1 + x^2)dy = 0$; when $x = 0, y = 2$.
	5.	Solve the D.E.: (i) $(x^2 - xy + y^2)dx = xy dy$. (ii) $(y + \sqrt{x^2 + y^2})dx - x dy = 0$; when $x = \sqrt{3}, y = 1$. (iii) $(x^3 + 3xy^2)dx + (3x^2y + y^3)dy = 0$. (iv) $(2y^2 + 3xy - 2y + 6x)dx + x(x + 2y - 1)dy = 0$. (v) $\frac{dy}{dx} + 2xy \tan x^2 = 2x \sec x^2$.

		(vi) $x \frac{dy}{dx} - y = x^k y^n$, where $n \neq 1$ and $k + n \neq 1$.
03	1.	Form the differential equation by eliminating arbitrary constant(s) of the equation: $y = c_1 e^x + c_2 e^{-x}$.
	2.	Obtain the differential equation of the family of straight lines with slope and y -intercept equal.
	3.	Reduce the following differential equation to a variables separable equation and hence solve it: $(x + y)(dx - dy) = dx + dy$.
	4.	Solve the initial value problems (IVP): $(3x + 8)(y^2 + 4)dx + 4y(x^2 + 5x + 6)dy = 0$, $y(1) = 3$.
	5.	Solve the D.E.: (i) $xy dx - (x^2 + 3y^2) dy = 0$. (ii) $(y - \sqrt{x^2 + y^2}) dx - x dy = 0$; when $x = 0, y = 1$. (iii) $x(x^2 + y^2 - a^2) dx + y(x^2 - y^2 - b^2) dy = 0$. (iv) $2y(x^2 - y + x) dx + (x^2 - 2y) dy = 0$. (v) $y \frac{dx}{dy} + x = y^2 + 3y + 2$. (vi) $y(6y^2 - x - 1) dx + 2x dy = 0$.
04	1.	Form the differential equation by eliminating arbitrary constant(s) of the equation: $y = c_1 + c_2 e^{3x}$.
	2.	Obtain the differential equation of the family of straight lines with algebraic sum of the intercepts fixed as α .
	3.	Reduce the following differential equation to a variables separable equation and hence solve it: $(x + y + 1)dx - (2x + 2y + 1)dy = 0$.
	4.	Solve the initial value problem that consist of the differential equation $x \sin y dx + (x^2 + 1) \cos y dy = 0$ and the initial condition $y(1) = \frac{\pi}{2}$.
	5.	Solve the D.E.: (i) $y dx = (x + \sqrt{y^2 - x^2}) dy$. (ii) $(y + \sqrt{x^2 + y^2}) dx - x dy = 0$; when $y(1) = 0$. (iii) $(y^2 - 2xy + 6x) dx - (x^2 - 2xy + 2) dy = 0$. (iv) $(x^2 + y^2 + 1) dx + x(x - 2y) dy = 0$. (v) $(1 - x^2) \frac{dy}{dx} + 2xy = (1 - x^2)^2$. (vi) $\frac{dy}{dx} + \frac{y}{2x} = \frac{x}{y^3}$, when $y(1) = 2$.
05	1.	Form the differential equation by eliminating arbitrary constant(s) of the equation: $y = c_1 e^{2x} + c_2 e^{3x}$.
	2.	Obtain the differential equation of the family of circles with center on the x -axis.
	3.	Reduce the following differential equation to a variables separable equation and hence solve it: $(x - y)^2 \frac{dy}{dx} = a^2$.
	4.	Solve the initial value problems (IVP): $(3x + 8)(y^2 + 4)dx - 4y(x^2 + 5x + 6)dy = 0$, $y(1) = 2$.
	5.	Solve the D.E.:

		(i) $x dy - y dx - \sqrt{x^2 - y^2} dx = 0$. (ii) $y^2 dx + (x^2 + 3xy + 4y^2) dy = 0$; when $y(2) = 1$. (iii) $(2x^3 + 3y) dx + (3x + y - 1) dy = 0$. (iv) $(3x^2 y^4 + 2xy) dx + (2x^3 y^3 - x^2) dy = 0$. (v) $(1 + x) \frac{dy}{dx} + y = 2\sqrt{x}$. (vi) $2x^3 \frac{dy}{dx} = y(y^2 + 3x^2)$.
06	1.	Form the differential equation by eliminating arbitrary constant(s) of the equation: $y = x + c_1 e^x + c_2 e^{-x}$.
	2.	Obtain the differential equation of the family of circles with fixed radius r and tangent to the x -axis.
	3.	Reduce the following differential equation to a variables separable equation and hence solve it: $\frac{dy}{dx} = \sin(x + y) + \cos(x + y)$.
	4.	Solve the initial value problems (IVP): $(y + 2) dx + y(x + 4) dy = 0$, $y(-3) = -1$.
	5.	Solve the D.E.: (i) $2x dy - 2y dx - \sqrt{x^2 + 4y^2} dx = 0$. (ii) $(16x + 5y) dx + (3x + y) dy = 0$; the curve pass through the point $(1, -3)$. (iii) $(6x + y^2) dx + y(2x - 3y) dy = 0$. (iv) $y(x + y) dx + (x + 2y - 1) dy = 0$. (v) $\frac{dy}{dx} + \frac{2xy}{1 + x^2} = \frac{e^x}{1 + x^2}$. (vi) $dy + (4y - 8y^{-3}) x dx = 0$.
07	1.	Form the differential equation by eliminating arbitrary constant(s) of the equation: $y = c_1 e^x + c_2 x e^x$.
	2.	Obtain the differential equation of the family of parabolas with vertex and focus on the x -axis.
	3.	Reduce the following differential equation to a variables separable equation and hence solve it: $\frac{dy}{dx} = (4x + y + 1)^2$.
	4.	Solve the initial value problems (IVP): $2x(y + 1) dx - y dy = 0$; when $x = 0$ and $y = -2$.
	5.	Solve the D.E.: (i) $t(s^2 + t^2) ds - s(s^2 - t^2) dt = 0$. (ii) $x dy - y dx = (x^2 + y^2) dx$; when $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$. (iii) $(x^2 - 2xy - y^2) dx - (x + y)^2 dy = 0$. (iv) $y(x + y + 1) dx + x(x + 3y + 2) dy = 0$. (v) $\frac{dx}{dy} + 2xy = 2e^{-y^2}$. (vi) $\frac{dy}{dx} (x^2 y^3 + xy) = 1$.
08	1.	Form the differential equation by eliminating arbitrary constant(s) of the equation:

	$y = c_1 e^{2x} \cos 3x + c_2 e^{2x} \sin 3x .$
2.	Obtain the differential equation of the family of parabolas with axis parallel to the x -axis.
3.	Reduce the following differential equation to a variables separable equation and hence solve it: $\frac{dy}{dx} = (x + y)^2 .$
4.	Solve the initial value problems (IVP): $(1 + y^2)dx + (1 + x^2)dy = 0$; when $x = 0, y = -1 .$
5.	Solve the D.E.: (i) $(x^3 + y^2 \sqrt{x^2 + y^2})dx - xy \sqrt{x^2 + y^2} dy = 0 .$ (ii) $(x - y)dx + (3x + y)dy = 0$; when $x = 3, y = -2 .$ (iii) $(2xy - 3x^2)dx + (x^2 + 2y)dy = 0 .$ (iv) $(4xy + 3y^2 - x)dx + x(x + 2y)dy = 0$ (v) $\frac{dy}{dx} - \frac{2x+1}{x^2} y = 1 .$ (vi) $xy - \frac{dy}{dx} = y^3 e^{-x^2} .$